Strategy Land

Richard Rumelt's Web Journal on Good and Bad Strategy in Business, Politics, and Economics

Learning and Market-Share

The spurious relationship between market share and profitability

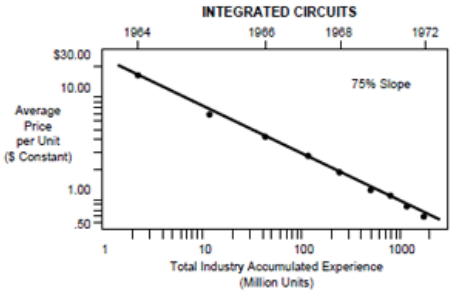
Part 2 of What Keeps High Profits High? [1 **2** 3 4 5]

February 18, 2010 By Richard Rumelt

The strategy field's first systematic theory about these sustained differences in performance attributed them to learning and market-share. Indeed, during the late 1970s, almost every book, paper, and address on business strategy put learning and market-share on center stage. Today we know that this early focus on learning and market-share grossly overstated their importance. That understanding was the hard-won result of a number of complex research efforts.

If one firm is more successful than another, there must be something that prevents the follower from catching up. I have given the name isolating mechanisms[1] to whatever it is that sustains efficiency differences among firms. The first isolating mechanism to gain widespread attention by strategists was learning-by-doing, or more specifically, the experience curve as popularized by the Boston Consulting Group.

Everyone knows that you get better at something after doing it a number of times. This kind of learning-by-doing was thrust into the economics literature by Wright's 1936 paper on airframe production.[2] Later research[3] showed the same effects during the war-time production of B-17s at Boeing's No. 2 plant. The first batch of aircraft, assembled in 1941, took about 140,000 labor hours each. A year later, the same aircraft were taking only 45,000 labor hours to assemble, and labor hours per aircraft dropped to 15,000 by the time the model was discontinued in 1945.



I saw my first experience curve in Bruce Henderson's office at Boston Safe Deposit and Trust Company in 1966. Henderson led the bank's consulting arm, named "The Boston Consulting Group," and had analyzed cost data from client Texas Instruments' semiconductor business. The log-log plot showed unit costs falling as cumulative production rose. My reaction was interest, but not the excitement Henderson was looking for. I went back to my doctoral studies and he went forward to build a major new consulting firm around the learning-curve, or what he called the *experience-curve*. Henderson's argument was that the experience-curve meant that once a firm got into the lead, it could stay there. It offered an explanation for sustained success under competition.

If accumulated experience reduces costs, then the firm with the most experience should be the low-cost market leader. And, having the most experience should usually translate into having more market share. In the mid-1970s, strategy consulting split between those pointing to cumulative experience versus those pointing at market share as the chief driver of sustained high profitability.

On of the most influential arguments about the value of market share came from a 1975 *Harvard Business Review* article "Market Share–a Key to Profitability."[4] The article reported that "There is no

doubt that market share and return on investment are strongly related. Exhibit I shows average pretax ROI figures for groups of businesses in the PIMS project that have successively increasing shares of their markets." From the viewpoint of market-share proponents, gaining share was a form of investment. For example, economist William Shepherd[5] claimed that a firm "can invest present profits in building up a higher future market share."

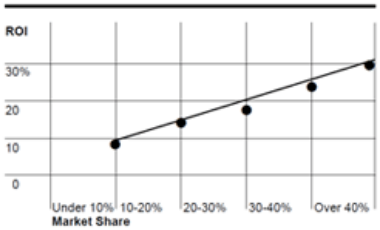
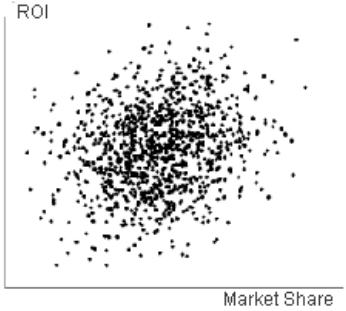


Figure I From "Market Share--a Key to Profitability"

In the late 1970s, Robin Wensley[6] and I were fascinated by the argument that a business' profitability was directly connected to its market share. It seemed like each month another article or book appeared pushing the "market-share" point of view and illustrated it with a bar chart or line graph like that of Figure I (above). We dug into the PIMS database being used in most of this research and were initially taken aback by the huge amount of variation not shown on such bar and line chart summaries. The adjoining chart shows actual raw data on market-share and ROI, each data point being one business. Consultants would not capture many clients with this splatter, but this is what real economic data always looks like. Although it is hardly striking to the eye, in this scatter market share does explain about 8% of the variance in ROI. You can see the association as a small relative absence of data points in the upper left and lower right quadrants.



ROI versus Market Share--Real Data

Despite the variance in the data, we set out to measure the cost of gaining market share. The idea was simple: If market share were valuable, then in a competitive economy it must cost something to grab it against competitors. We wanted to compare the cost of gaining share to the subsequent speed of share decay. We expected share to be more costly in industries where it was "sticky" and cheaper in industries where such gains were quickly erased.

To our surprise, we could not find a cost to gaining share. Looking over a five-year window of time, the faster a business gained share the faster its profitability increased. This result flew in the face of the idea that chunks of market-share are valuable resources, like plots of land. If market share were intrinsically valuable, rivals would not give it up without fighting back, imposing costs on the business that is trying to gain share.

If market-share wasn't valuable, then why was there a share-ROI association in the data? One explanation was that the association was spurious. That is, the relationship between share and ROI was not causal. Instead, both ROI and share were "caused" by something else, something not being measured in the data. There is, for instance, a statistical association across households between size of house and quality of car. That association exists there because people with higher incomes buy larger homes and also buy better automobiles. Buying a more expensive car will not increase your living space!

Instead of market-share "causing" higher ROI, we hypothesized that both share and ROI were both indicators of *success*. Companies with successful products or marketing campaigns had both higher market-shares and higher ROI. The market-share crowd was mistaking an association for a cause.

Robin Wensley and I tested this idea statistically by breaking changes in share into predicted and unpredicted components. The prediction was based on changes in price, marketing, and R&D, as well as past performance. The data showed that market share did respond to changes in price, quality, marketing and research. For example, price cuts pushed it up and price cuts by competitors pushed it down. So, there was a clear cost of gaining market share—the costs of lower prices, higher marketing and higher R&D. But, that predictable part of the change in market share, the part that was paid for, had *no association* with changes in profitability. Rather, it was the unpredicted part of the change in market share that boosted profitability. We concluded that the apparent connection between profitability and market-share was spurious, being due to a third hidden process—good luck or good management.[7]

If market share is not strongly implicated as an independent driver of profitability, what happens to the experience-curve school of thought? Today, with hindsight, one sees that the curve Henderson showed me was just basic semiconductor economics. With a given production set-up, unit costs fall as yield (the fraction of good parts) rises. Yield, in turn, rises as sources of defects are identified and corrected. Much more importantly, over time, producers were able to pack more and more transistors onto each square millimeter of silicon—the process now dubbed "Moore's Law."

But the critical element, one missing from Henderson's presentation, was that Moore's Law applies to the industry as a whole, not the individual competitor. Although Intel managers love to point at how Moore's Law allows each new generation of processors to be faster and more complex, the same "Law" helps along most competitors as well. Henderson's client, Texas Instruments, discovered this hard fact when, after racing down the hand-calculator "experience curve" in search of a protected position, it found, not a pot of gold, but a host of Taiwanese competitors with equivalent costs. Most of the cost reductions shown on Henderson's curve were not proprietary and could not be the source of sustained performance differences.

Another favorite example of the experience curve used by BCG in the early 1970s was of the automobile industry. The horizontal axis was cumulative automobiles produced by a company and the vertical axis was unit cost, both being log scales. Three large circles sat on the curve, each identified as one of the Big Three automakers. General Motors led, followed by Ford and Chrysler. The chart also showed the three firms' profit margins and the text argued that as long as General Motors retained its market share lead, its faster accumulating experience would keep it ahead on share, cost, and profit. By the end of the 1970s, the chart was unusable because Toyota and Honda had jumped into the lead in efficiency and quality. These new competitors had confounded the economists' talk of "entry barriers" and the strategy community's talk of "experience effects." Not only were they better at designing and making automobiles, the Big Three could not seem to replicate their methods after a decade or more of careful study and research.

Footnotes

1. See Rumelt, Richard P., "Towards a Strategic Theory of the Firm," in Foss, Nicolai J. (ed.), *Resources Firms and Strategies: A Reader in the Resource-Based Perspective,* Oxford University Press, 1997. pp. 131-145. This article first appeared in Lamb, Robert, *Competitive Strategic Management*, Prentice-Hall, 1984, 556-70. [↩]

2. Wright, T. P. "Factors Affecting the Cost of Airplanes," *J. Aeronautical Science*, 1936, 3, p. 122-128 [↩]

3. Alchain, Armen, "Reliability of Progress Curves in Airframe Production," *Econometrica*, 1963, 31, pp. 679-94. [↩]

4. Buzzell, Robert D., Gale3, Bardley T., and Ralph G.M. Sultan, "Market Share–a Key to Profitability," *Harvard Business Review*, Jan-Feb 1975:97-105. [↩]

5. Shepherd, W.G., "The elements of market structure," *Review of Economics and Statistics*, 54, 1972: 25-37. [↩]

6. Professor of Policy & Marketing, Warwick Business School, UK. [↩]

7. Rumelt, Richard and Robin Wensley, "In Search of the Market Share Effect." *Proceedings of the Academy of Management*, August 1981, p. 1-5. [↩]

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